

R and Python package for REDCap API (beginner level)

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REDCapCon September 2025

- Shawn Garbett, Vanderbilt University, Dept of Biostatistics
- Günther Reznicek, Ruhr-Universität Bochum, Dept of Obstetrics and Gynecology
- Will Beasley, University of Oklahoma, Office of Clinical Research Informatics

Follow along with us at <https://github.com/OuhscBbmc/redcapcon-2025-r-api>

1 Agenda

Total time: **70m**

1. Introductions 1m
2. Motivation (Günther) 5m
 1. vs csv
 2. vs writing your own R code with httr/RCurl
 3. Reproducibility!
3. Prereqs (Günther)
 1. REDCap project exists
 2. user access to
 3. dev machine has R & redcapAPI/REDCapR
4. Basic Introduction redcapAPI/REDCapR (sg) 10m (unstructured)
5. Security (sg) 5m (shelter)
6. redcapAPI (sg) – Type Theory (type casting) REDCapR/redcapAPI
7. Longitudinal (Will)
 1. Roll your own with redcapAPI/REDCapR
 2. REDCapTideR
8. Writing (brief -Will)
9. REDCap API 2.0 (Günther)
10. Further Resources (Will)
 1. Community is for REDCap admins, which many biostatisticians are not
 2. REDCap Tools

3. Stack Overflow redcap tag
4. Reddit ProjectREDCap subreddit
5. Python and Ruby packages.
11. Closing
 1. API & Human Connections
 2. RAISE
 3. Security Committee
 4. Questions?

2 Motivation - Working with REDCap Data

- REDCap is widely used for data collection in clinical & research contexts
- Once collected, the real work begins: **analysis**
- Key question: *How do you usually get your data out of REDCap?*

2.1 CSV Export vs API Access - Two Different Worlds

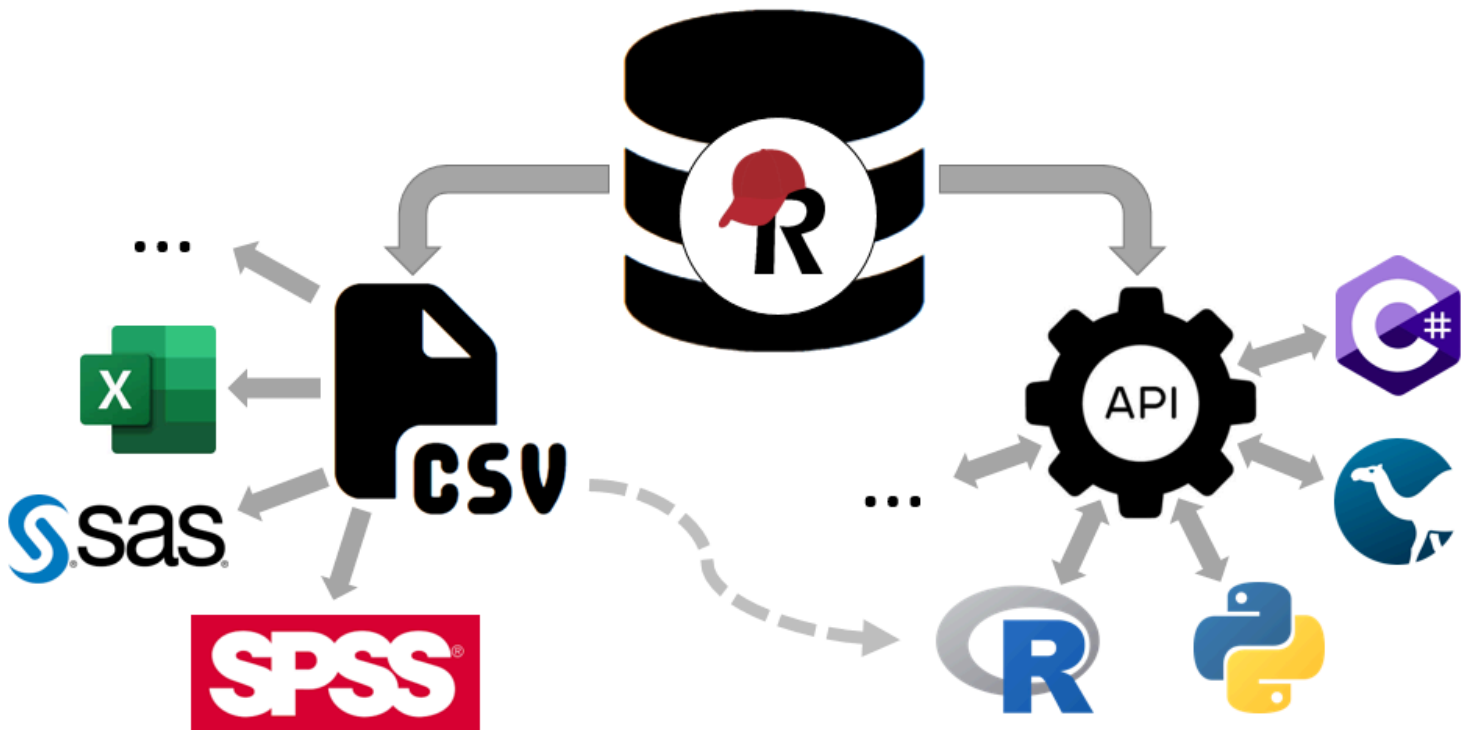


Figure 1: REDCap Data? CSV or API

CSV Export

- Manual download from REDCap
- Creates a **static snapshot** of the data
- Can include **all data** or a defined subset (report)
- Good for **one-time analyses** or data sharing
- No programming skills required
- Easy to open in Excel, SPSS, SAS, ...

API Access

- Connects directly to the REDCap project
- Always retrieves the **latest version** of the data
- Integrates into **R/Python scripts and workflows**
- Can also export reports programmatically
- Automates repetitive steps and reduces errors

- Ensures **reproducibility** through scripted pipelines

Mixed / Hybrid Approach

- **Exported CSVs** can be used as input for scripted pipelines
- Allows combining manual exports with automated analysis steps
- Reports can be exported both manually or via the API → more control from REDCap side
- Programmatic pipelines still ensure reproducibility

2.2 The Concept of Living (Meta)Data

- REDCap projects **evolve continuously**:
 - New participants enrolled
 - Existing records updated
 - Data quality issues corrected*
 - **New fields or forms added** over time
- **CSV Export**
 - Provides only a **static snapshot** of data and metadata
 - Must be re-downloaded whenever project structure or data changes
 - Risk of outdated or inconsistent files
- **API Access**
 - Always pulls the latest data and metadata
 - Integrates directly into scripted workflows**
 - Supports reproducibility and automation even as the project changes**
- These issues could very well be detected through analyses *outside* of REDCap
 - Of course, metadata changes may required the scripted pipeline to be adapted

2.3 Why do we care?

Reproducibility & Transparency

- Data analyses should be:
 - *Reproducible*: same script = same results, anytime
 - *Transparent*: clear documentation of steps
- Using R scripts with API pulls:
 - analysis becomes a documented pipeline,
 - easier to share with collaborators,
 - simpler to audit or reproduce months later.

Quality & Efficiency Benefits

- Automated API workflows reduce risk of:
 - Using the wrong/outdated CSV version
 - Manual copy-paste errors
- Type consistency & controlled variable names directly from Codebook/API.
- Early data quality checks possible (missing data, unexpected values, etc.).

Thus, while CSV exports will always remain a valuable option, especially for quick checks, simple analyses, or collaboration with colleagues who prefer spreadsheets, they quickly reach their limits in projects where data and metadata keep evolving and frequent analyses need to be performed.

REDCap projects can be considered “living systems”: both, data and metadata change over time. To keep up with these changes, reproducibility and automation become essential. Programmatic access via the API provides a reliable way to ensure that both data and metadata are always current, and that every step of the analysis can be repeated without guesswork.

3 Accessing REDCap Data via the API: Prerequisites

3.1 REDCap Projects and Data Organization

- **Fields** (Variables)
 - Basic units of data (e.g., age, blood pressure, diagnosis)
 - Each field has a **type** (text, number, date, multiple choice, etc.)
- **Instruments** (Forms)
 - Simply a **collection of fields**
 - Used for data entry or surveys
- **Classic projects**
 - One timeline (one “arm” with one event; not visible in the UI)
 - Data organized by instruments
 - Instruments can be **repeatable**
- **Longitudinal projects**
 - One or more arms with **events** (timepoints)
 - Instruments assigned to one or multiple events
 - Instruments/events can be **repeatable** (but instruments cannot repeat on repeating events)
- **For data analysts**
 - Understanding project structure is essential
 - Access may be **via API key** without REDCap login

Classic		Longitudinal						
Data Collection Instrument	Status	Data Collection Instrument	Initial Data	Week 1	Week 2	Week 3	Week 4	Final Data
Demographics		Pre-Screening Survey (survey)						
Baseline Data		Participant Info Survey (survey)						
Month 1 Data		Participant Morale Questionnaire (survey)						
Month 2 Data		Completion Data (to be entered by study personnel only)						
Month 3 Data		Communication Log						
Completion Data								

Figure 2: Classic vs Longitudinal Projects

3.2 Metadata: Data Dictionary and Codebook

- **Data Dictionary**
 - CSV file describing all fields in a project
 - Includes field names, types, choices, validation rules
 - Downloadable from project setup page (requires *Project Design and Setup* right) or **API** (“Export Metadata”)
- **Codebook**
 - Rich, human-readable view of project metadata
 - Shows instruments, events (in longitudinal projects), and repeatability
 - Includes list of missing data codes, if defined
 - Features: search within REDCap, or save/export as PDF for offline use
 - The results of *several* API methods have to be combined to obtain the information contained in the Codebook (e.g., “Export Project Info” for missing data codes)
- **Why is this important for analysts?**
 - Metadata defines the **meaning of data fields**, including coding of categorical choices and labels or annotations that specify units for numerical data
 - Essential for data quality checks and interim analyses
 - Needed to keep up with changes in “living (meta)data”

Data Dictionary, viewed in VS Code

```

1  "Variable / Field Name","Form Name","Section Header","Field Type","Field Label","Choices,
Calculations, OR Slider Labels","Field Note","Text Validation Type OR Show Slider Number","Text
Validation Min","Text Validation Max",Identifier?,"Branching Logic (Show field only if...)",
"Required Field?","Custom Alignment","Question Number (surveys only)","Matrix Group Name",
"Matrix Ranking?","Field Annotation"
2  participant_id,prescreening_survey,,text,"Participant ID",,,,,,,,,,
3  dob,prescreening_survey,"Please fill out the information below.",text,"Date of birth",,,
date_ymd,,,,,,,,,
4  email,prescreening_survey,,text,"E-mail address",,,email,,y,,,,,,,,,
5  has_diabetes,prescreening_survey,,truefalse,"I currently have Type 2 Diabetes",,,,,,,,,,
6  consent,prescreening_survey,,checkbox,"By checking this box, I certify that I am at least 18
years old and that I give my consent freely to participate in this study.",,"1, I
consent",,,,,,,,,,
7  first_name,participant_info_survey,"As a participant in this study, please answer the questions
below. Thank you!",text,"First Name",,,,,y,,,,,,,,,
8  last_name,participant_info_survey,,text,"Last Name",,,,,y,,,,,,,,,
9  address,participant_info_survey,,notes,"Street, City, State, ZIP",,,,,y,,,,,,,,,
10 telephone_1,participant_info_survey,,text,"Phone number",,"Include Area Code",phone,,y,,,,,,,,,
11 ethnicity,participant_info_survey,,radio,Ethnicity,"0, Hispanic or Latino | 1, NOT Hispanic or
Latino | 2, Unknown / Not Reported",,,,,,,,,LH,,,,,
12 race,participant_info_survey,,dropdown,Race,"0, American Indian/Alaska Native | 1, Asian | 2,
Native Hawaiian or Other Pacific Islander | 3, Black or African American | 4, White | 5, More
Than One Race | 6, Unknown / Not Reported",,,,,,,,,,
13 sex,participant_info_survey,,radio,Sex,"0, Female | 1, Male | 2, Other | 3, Prefer not to
say",,,,,,,,,,
14 height,participant_info_survey,,text,"Height (cm)",,,number,130,215,,,,,,,,,
15 weight,participant_info_survey,,text,"Weight (kilograms)",,,integer,35,200,,,,,,,,,
16 age,participant_info_survey,,calc,Age,"round(datediff([initial_data_arm_1]
[survey-date-completed:prescreening_survey], [initial_data_arm_1][dob], ""y""),1)",,,,,,,,,,
@HIDDEN-SURVEY"

```

Figure 3: Data Dictionary CSV, viewed in VS Code

Codebook (viewed inside REDCap)

Field finder: — When viewing this page, collapse: ☐ all fields ☐ all additional tables

Instruments			Events		
Instrument	Form Name	Events	Event Name	Unique event name	Event ID
Pre-Screening Survey	prescreening_survey	initial_data_arm_1	Initial Data	initial_data_arm_1	1103
Participant Info Survey	participant_info_survey	initial_data_arm_1	Week 1	week_1_arm_1	1104
Participant Morale Questionnaire	participant_morale_questionnaire	week_1_arm_1	Week 2	week_2_arm_1	1105
		week_2_arm_1	Week 3	week_3_arm_1	1106
		week_3_arm_1	Week 4	week_4_arm_1	1107
		week_4_arm_1	Final Data	final_data_arm_1	1108
Completion Data (to be entered by study personnel only)	completion_data	final_data_arm_1			
Communication Log	communication_log	initial_data_arm_1			

Codes for Missing Data	
Code / Value	Label
NI	No information
UNK	Unknown
NASK	Not asked
OTH	Other

Collapse all instruments

#	Variable / Field Name	Field Label <i>Field Note</i>	Field Attributes (Field Type, Validation, Choices, Calculations, etc.)
Instrument: Pre-Screening Survey (prescreening_survey) Enabled as survey Collapse			
	1	[participant_id]	Participant ID
	2	[dob]	Section Header: Please fill out the information below. Date of birth
	3	[email]	E-mail address
	4	[has_diabetes]	I currently have Type 2 Diabetes
	5	[consent]	By checking this box, I certify that I am at least 18 years old and that I give my consent freely to participate in this study.

Figure 4: Codebook viewed inside REDCap

3.3 User Rights – Viewing vs Export Rights

- **Viewing Rights**
 - Apply when logged into REDCap’s web interface
 - Define which records and fields a user can see on forms or reports
- **Export Rights**
 - Apply when downloading data or using the API
 - Control the **level of detail** that can leave REDCap

Export Rights Levels

- **No Access** → user cannot export data
- **De-Identified** → removes direct identifiers, all dates, and all notes and unvalidated text fields

- **Remove All Identifier Fields** → strips fields marked as identifiers
- **Full Data Set** → no restrictions

A note on **Data Access Groups**: Users who are in a DAG only ever can access records in that DAG. Data analysts (usually) need to get data from all DAGs, and thus they are (usually) not in a DAG.

The screenshot below shows the part of the User Rights dialog that sets data viewing and export rights. As shown, these might be the rights assigned to a statistician who does not normally log into REDCap but accesses data using the API only.

Privileges for Viewing and Exporting Data									
Data Viewing Rights pertain to a user's ability to view or edit data on pages in the project (e.g., data entry forms, reports). Users with 'No Access' Data Viewing Rights for a given instrument will not be able to view that instrument for any record, nor will they be able to view fields from that instrument on a report. Data Export Rights pertain to a user's ability to export data from the project, whether through the Data Exports page, API, Mobile App, or in PDFs of instruments containing record data. Note: Data Viewing Rights and Data Export Rights are completely separate and do not impact one another.									
	Data Viewing Rights					Data Export Rights			
	No Access (Hidden)	Read Only	View & Edit	Delete	Edit Survey Responses	No Access	De-Identified*	Remove All Identifier Fields	Full Data Set
Pre-Screening Survey (survey)	☒	☐	☐	☐	☐	☐	☒	☐	☐
Participant Info Survey (survey)	☒	☐	☐	☐	☐	☐	☒	☐	☐
Participant Morale Questionnaire (survey)	☒	☐	☐	☐	☐	☐	☒	☐	☐
Completion Data (to be entered by study personnel only)	☒	☐	☐	☐		☐	☒	☐	☐
Communication Log	☒	☐	☐	☐		☐	☒	☐	☐

* De-identified means that all free-form text fields will be removed, as well as any date/time fields and Identifier fields.

Figure 5: privileges

3.4 Reports in REDCap

- **Reports = custom data views**
 - Define subsets of fields (and records, when filtering)
 - Useful for scoped analyses where not all data is required
- **Access considerations**
 - Reports respect user **viewing** rights when viewed inside REDCap
 - Reports respect **export** rights when exporting the data
 - Access to a specific report can be limited to certain users/roles in the report settings
 - Data Access Group restrictions are always observed
- **Exports**
 - Reports can be downloaded as CSV
 - Reports can also be exported programmatically via the **API** ("Export Reports")

Given the user rights shown above, a user viewing a report while logged into REDCap will no see any data:

Participant ID participant_id	Event Name redcap_event_name	Repeat Instrument redcap_repeat_instrument	Repeat Instance redcap_repeat_instance	Sex sex	Age age	Height (cm) height	Weight (kilograms) weight	On average, how many pills did you take each day last week? pmq1	Using the handout, which level of dependence do you feel you are currently at? pmq2	The choices you made choices
1	Initial Data			-	-	-	-	-	-	-
1	Week 1			-	-	-	-	-	-	-
1	Week 2			-	-	-	-	-	-	-
1	Week 3			-	-	-	-	-	-	-
1	Week 4			-	-	-	-	-	-	-
1	Final Data			-	-	-	-	-	-	-

Figure 6: report-viewed-in-redcap

However, that same report, exported as a CSV (either by downloading it, or by using the “Export Report” API method), will contain the data:

```

1 participant_id,redcap_event_name,redcap_repeat_instrument,redcap_repeat_instance,sex,age,height,weight,pmq1,pmq2,choices,life,job,family,participant_morale_questionnaire_complete,complete_study,withdraw_reason,discharge_summary_4,completion_data_complete
2 1,initial_data_arm_1,,1,25.6,130,169,,,,,,
3 1,week_1_arm_1,,,,,1,4,4,4,4,1,2,,,,
4 1,week_2_arm_1,,,,,2,1,3,2,1,4,2,,,,
5 1,week_3_arm_1,,,,,2,0,5,2,5,4,2,,,,
6 1,week_4_arm_1,,,,,1,2,5,5,1,4,2,,,,
7 1,final_data_arm_1,,,,,,0,2,1,2

```

Figure 7: report-exported-as-csv

Note: When exporting **de-identified** data from a report via CSV download, date field can either be removed or date-shifted. This option is not available for API exports, where date fields are always removed.

Date and datetime fields:

☒ Remove all date and datetime fields

— OR —

☐ Shift all dates by value between 0 and 364 days
(shifted amount determined by algorithm for each record)

[What is date shifting?](#)

☐ Also shift all survey completion timestamps by value between 0 and 364 days
(shifted amount determined by algorithm for each record)

Figure 8: export-remove-vs-dateshift

Below, line 7 of the CSV with date-shifted values is shown.

```

7 1,final_data_arm_1,,,,,,0,2025-05-06,2,2025-05-06,2025-05-06,1,2

```

Figure 9: report-with-date-shifting

Tip: Take this into account when planning an analyses that needs to take date differences into account and not using the **de-identified** export right is not an option *and* the export needs to be performed with the API. A workaround would be to calculate differences between dates in REDCap and exporting those instead of the actual date fields.

3.5 Introduction to the REDCap API

• What is the API?

- A set of endpoints to access REDCap data and metadata programmatically
- Enables integration with R, Python, and other tools

- **API User Rights**

- Separate from data export rights
- Key options:

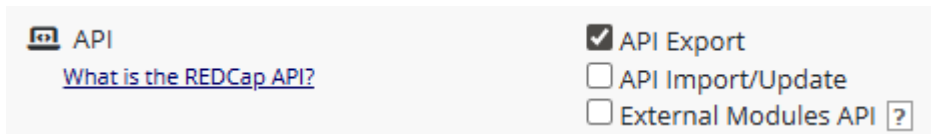


Figure 10: api-user-rights

- **Export** (read data/metadata)
 - **Import/Update** (write data)
 - **External Modules API** (access EM-provided methods)
- **API, API Playground, and API Documentation**

- Accessible from the *Applications* menu:

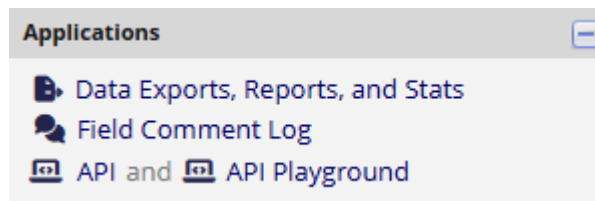


Figure 11: api-playground-menu

- Allows to request and then view/mangage the API token
- Show available methods
- Provide interactive examples for testing (read/write in development projects, read only in production projects)
- Provides concrete code snippets for various programming languages

API

The REDCap API is an interface that allows external applications to connect to REDCap remotely, and is used for programmatically retrieving or modifying data or settings within REDCap, such as performing automated data imports/exports from a specified REDCap project. For details on the capabilities of the REDCap API and how to use it, please see the [REDCap API documentation](#).

API Security: Best Practices

It is important to remember that when making requests to the REDCap API, you should always validate the REDCap server's SSL certificate to ensure the highest level of security during communication with the API. For details on what this means and how to do it, see the 'API Security: Best Practices' section in the [REDCap API documentation](#).

Obtain API token for "Accessing REDCap Data with the API"

Use the button below to request an API token for this project from your REDCap administrator. You will need a different token for each project you would like to access. Please note that your REDCap administrator is emailed every time a token is requested.

Request API token

Event names for Accessing REDCap Data with the API

Unique event name	Event Name	Arm
<code>initial_data_arm_1</code>	Initial Data	Arm 1
<code>week_1_arm_1</code>	Week 1	Arm 1
<code>week_2_arm_1</code>	Week 2	Arm 1
<code>week_3_arm_1</code>	Week 3	Arm 1
<code>week_4_arm_1</code>	Week 4	Arm 1
<code>final_data_arm_1</code>	Final Data	Arm 1

Once an API token has been created, it can be viewed and managed:

Your API token for project "Accessing REDCap Data with the API"

The API token below is **ONLY for you** and will work **ONLY with this project**. This token allows special access to REDCap data and **should NOT be shared with others**. If you think your token has been compromised, then please contact your REDCap administrator immediately *AND* either delete or regenerate your token by using the buttons below.

 **API Token:** 362FC0927A375904749A19E837A553F9



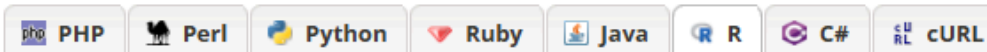
Delete token

Finished using the API for this project? If so, please delete your token for security reasons.

Regenerate token

Think someone else knows your token? If so, please regenerate your token for security reasons.

Interactive API method calls executed in the API Playground are shown as code:



Displayed in the box below is the code you would use to execute this API request in the selected programming language.

```
#!/usr/bin/env Rscript
token <- "362FC0927A375904749A19E837A553F9"
url <- "https://dev-redcap/api/"
formData <- list("token"=token,
  content='report',
  format='csv',
  report_id='1109',
  csvDelimiter=',',
  rawOrLabel='raw',
  rawOrLabelHeaders='raw',
  exportCheckboxLabel='false',
  returnFormat='json'
)
response <- http::POST(url, body = formData, encode = "form")
result <- http::content(response)
print(result)
```

- **API Tokens**

- Unique key linked to a specific **user and project**
- Tokens provide both,
 - **Authentication** (like username + password) and
 - **Authorization** (role/user rights of the user in that specific project)
- REDCap provides tools for token management (cycling, revocation, transfer)
- Must be kept **secure** (details on token handling *outside of REDCap* will follow later)

Tip: If you are a data analyst without access to the production REDCap project but need to learn how the REDCap API works, ask for access to a demo or test project. This allows you to safely “play around” and practice API calls without risking production data.

3.6 Prerequisites on the Analyst’s Side (R)

- **R Installation**

- Install the latest version of R (and optionally RStudio IDE)

- **Network Connectivity**

- Ensure your machine can reach the REDCap API endpoint
- Example: `https://redcap.domain.name/api/`

- **API Token**

- Personal key tied to user + project
- Required for all API calls

- **R Packages**

- While not strictly required, install packages that abstract away a lot of the low level REDCap API interactions and data transformations, such as, e.g.,
 - redcapAPI
 - REDCapR
 - REDCapTidieR

- **Recommended IDE Enhancements**

- Code completion and linting (e.g., languageserver, lintr)
- Syntax highlighting for JSON/CSV
- Git integration for version control
- Notebook-style workflow (e.g., R Markdown or Quarto) for combining code, results, and documentation

With these prerequisites in place — understanding how REDCap organizes its data and metadata, knowing how user rights shape what can be exported, and setting up your own R environment with the necessary tools — you are ready to move from theory to practice. In the next section, we will take a closer look at the available R packages for interacting with the REDCap API and see how they can streamline your analysis workflows.

4 Basic Introduction redcapAPI/REDCapR

One has a choice of REDCapR or redcapAPI and both offer a rich set of features. The REDCapR package has better tibble integration with the more modern tidyverse style of operations. The redcapAPI package is targeted towards base R and includes additional support for automated environments. This introduction will cover examples using both libraries.

We begin with loading libraries and a token and a uri. NOTE: writing a token in R code is poor security practice. More on proper security in the next section.

```
uri      <- "https://redcap-dev-2.ouhsc.edu/redcap/api/"
token    <- "9A068C425B1341D69E83064A2D273A70"
```

5 Read Data (Unstructured)

5.1 REDCapR

```
rcr_1 <- REDCapR::redcap_read(redcap_uri=uri, token=token)
#> 24 variable metadata records were read from REDCap in 0.3 seconds. The http status code was 200.
#> The data dictionary describing 17 fields was read from REDCap in 0.2 seconds. The http status
code was 200.
#> 3 instrument metadata records were read from REDCap in 0.1 seconds. The http status code was 200.
#> 1 rows were read from REDCap in 0.1 seconds. The http status code was 200.
#> 2 data access groups were read from REDCap in 0.1 seconds. The http status code was 200.
#> 5 records and 1 columns were read from REDCap in 0.5 seconds. The http status code was 200.
#> Starting to read 5 records at 2025-09-09 18:08:59.403296.
#> Reading batch 1 of 1, with subjects 1 through 5 (ie, 5 unique subject records).
#> 5 records and 25 columns were read from REDCap in 0.3 seconds. The http status code was 200.
#>
rcr_1 <- rcr_1$data # Extract just the data
```

At this point, the data.frame rcr_1 has everything one needs to start analyzing the project.

```
rcr_1
#> # A tibble: 5 × 25
#>   record_id name_first name_last address telephone email dob age sex
#>   <dbl> <chr> <chr> <chr> <chr> <chr> <date> <dbl> <dbl>
#> 1 1 Nutmeg Nutmouse "14 Ros\u2026 (405) 32\u2026 nutt\u2026 2003-08-30 11 0
#> 2 2 Tumtum Nutmouse "14 Ros\u2026 (405) 32\u2026 tumm\u2026 2003-03-10 11 1
#> 3 3 Marcus Wood "243 Hi\u2026 (405) 32\u2026 mw@m\u2026 1934-04-09 80 1
#> 4 4 Trudy DAG "342 El\u2026 (405) 32\u2026 pero\u2026 1952-11-02 61 0
#> 5 5 John Lee Walker "Hotel \u2026 (405) 32\u2026 left\u2026 1955-04-15 59 1
#> # \u2026139 16 more variables: demographics_complete <dbl>, height <dbl>, weight <dbl>,
#> # bmi <dbl>, comments <chr>, mugshot <chr>, health_complete <dbl>,
#> # race__1 <dbl>, race__2 <dbl>, race__3 <dbl>, race__4 <dbl>,
#> # race__5 <dbl>, race__6 <dbl>, ethnicity <dbl>, interpreter_needed <dbl>,
#> # race_and_ethnicity_complete <dbl>

hist(rcr_1$weight)
```

Histogram of rcr_1\$weight

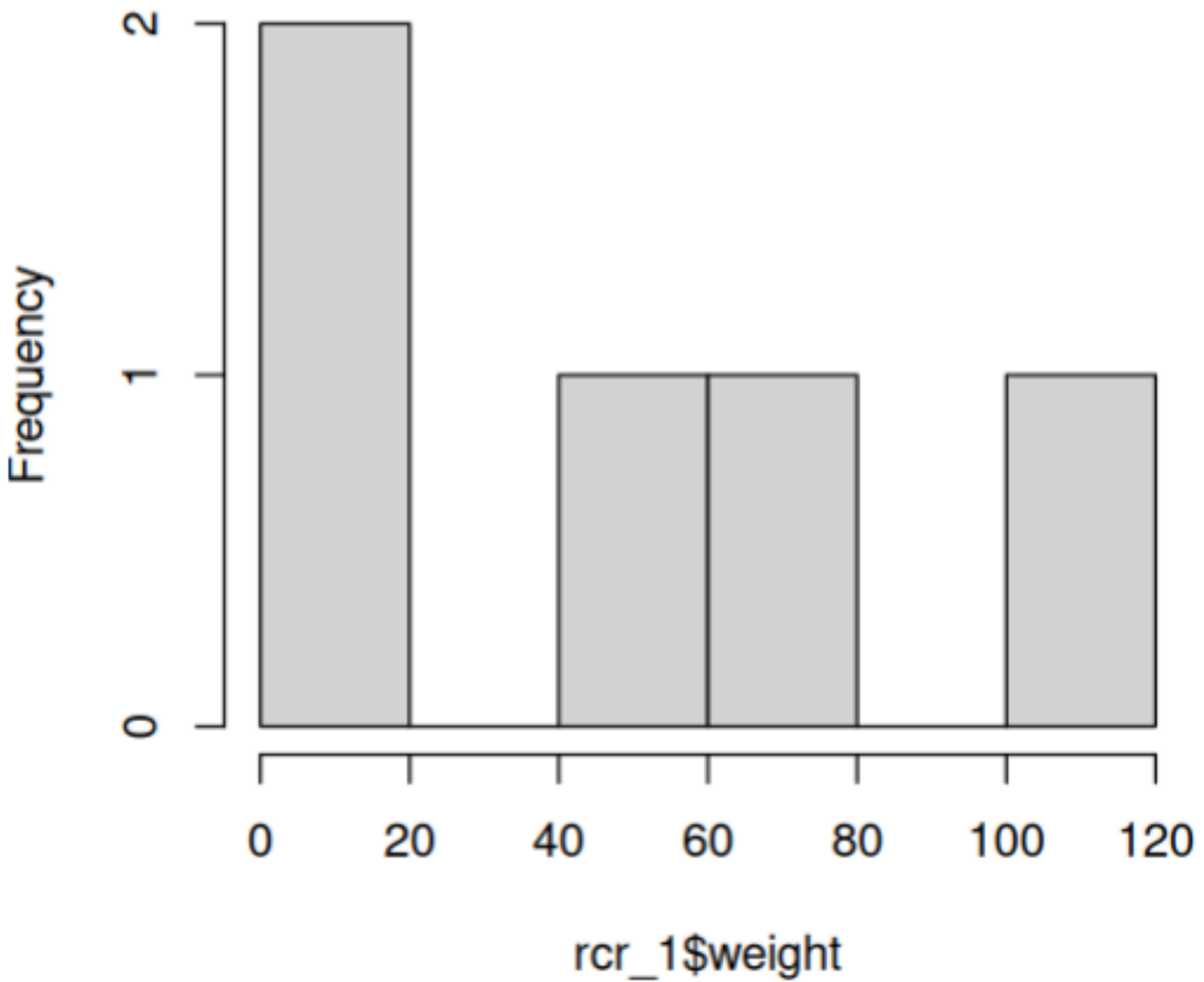


Figure 12: histogram-weight

```
summary(rcr_1)
#>   record_id  name_first      name_last      address
#>   Min.      :1    Length:5      Length:5      Length:5
#>   1st Qu.:2    Class :character  Class :character  Class :character
#>   Median :3    Mode  :character  Mode  :character  Mode  :character
#>   Mean      :3
#>   3rd Qu.:4
#>   Max.      :5
#>
#>   telephone      email      dob      age
#>   Length:5      Length:5      Min.    :1934-04-09  Min.    :11.0
#>   Class :character  Class :character  1st Qu.:1952-11-02  1st Qu.:11.0
#>   Mode  :character  Mode  :character  Median :1955-04-15  Median :59.0
#>                                     Mean  :1969-11-06  Mean  :44.4
```

```

#>               3rd Qu.:2003-03-10   3rd Qu.:61.0
#>               Max.      :2003-08-30   Max.      :80.0
#>
#>      sex      demographics_complete      height      weight
#> Min.      :0.0   Min.      :2           Min.      : 6.0   Min.      : 1
#> 1st Qu.:0.0   1st Qu.:2           1st Qu.: 7.0   1st Qu.: 1
#> Median :1.0   Median :2           Median :165.0   Median : 54
#> Mean      :0.6   Mean      :2           Mean      :110.2   Mean      : 48
#> 3rd Qu.:1.0   3rd Qu.:2           3rd Qu.:180.0   3rd Qu.: 80
#> Max.      :1.0   Max.      :2           Max.      :193.0   Max.      :104
#>
#>      bmi      comments      mugshot      health_complete
#> Min.      : 19.8   Length:5           Length:5           Min.      :0
#> 1st Qu.: 24.7   Class :character   Class :character   1st Qu.:0
#> Median : 27.9   Mode  :character   Mode  :character   Median :1
#> Mean      :110.9                                     Mean      :1
#> 3rd Qu.:204.1                                     3rd Qu.:2
#> Max.      :277.8                                     Max.      :2
#>
#>      race__1      race__2      race__3      race__4      race__5
#> Min.      :0.0   Min.      :0.0   Min.      :0.0   Min.      :0.0   Min.      :0.0
#> 1st Qu.:0.0   1st Qu.:0.0   1st Qu.:0.0   1st Qu.:0.0   1st Qu.:1.0
#> Median :0.0   Median :0.0   Median :0.0   Median :0.0   Median :1.0
#> Mean      :0.2   Mean      :0.2   Mean      :0.2   Mean      :0.2   Mean      :0.8
#> 3rd Qu.:0.0   3rd Qu.:0.0   3rd Qu.:0.0   3rd Qu.:0.0   3rd Qu.:1.0
#> Max.      :1.0   Max.      :1.0   Max.      :1.0   Max.      :1.0   Max.      :1.0
#>
#>      race__6      ethnicity interpreter_needed race_and_ethnicity_complete
#> Min.      :0.0   Min.      :0   Min.      :0.00   Min.      :0.0
#> 1st Qu.:0.0   1st Qu.:1   1st Qu.:0.00   1st Qu.:2.0
#> Median :0.0   Median :1   Median :0.00   Median :2.0
#> Mean      :0.2   Mean      :1   Mean      :0.25   Mean      :1.6
#> 3rd Qu.:0.0   3rd Qu.:1   3rd Qu.:0.25   3rd Qu.:2.0
#> Max.      :1.0   Max.      :2   Max.      :1.00   Max.      :2.0
#>
#>               NA's      :1

summary(lm(age ~ 1 + sex + bmi, data = rcr_1))
#>
#> Call:
#> lm(formula = age ~ 1 + sex + bmi, data = rcr_1)
#>
#> Residuals:
#>      1      2      3      4      5
#> -2.491  1.954  9.132  2.491 -11.086
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept) 63.34496    8.89980   7.118  0.0192 *
#> sex          13.55626    9.62958   1.408  0.2945
#> bmi         -0.24426    0.04337  -5.632  0.0301 *
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 10.55 on 2 degrees of freedom
#> Multiple R-squared:  0.9442, Adjusted R-squared:  0.8884
#> F-statistic: 16.92 on 2 and 2 DF, p-value: 0.05581

```

5.2 redcapAPI

redcapAPI has a different approach in that a connection object is created. This object maintains state locally for things like the data dictionary to minimize round trips to the server.

```
conn <- redcapAPI::redcapConnection(uri, token)
rca_1 <- exportRecordsTyped(conn)
```

```
head(rca_1)
#>   record_id name_first name_last          address      telephone
#>   email
#> 1         1   Nutmeg   Nutmouse 14 Rose Cottage St.\nKenning UK, 323232 (405) 321-1111
#>   nutty@mouse.com
#> 2         2   Tuntum   Nutmouse 14 Rose Cottage Blvd.\nKenning UK 34243 (405) 321-2222
#>   tummy@mouse.comm
#> 3         3   Marcus    Wood      243 Hill St.\nGuthrie OK 73402 (405) 321-3333
#>   mw@mwood.net
#> 4         4    Trudy     DAG      342 Elm\nDuncanville TX, 75116 (405) 321-4444
#>   peroxide@blonde.com
#> 5         5  John Lee   Walker    Hotel Suite\nNew Orleans LA, 70115 (405) 321-5555
#>   left@hippocket.com

hist(rca_1$weight)
```


Histogram of rcr_1\$weight

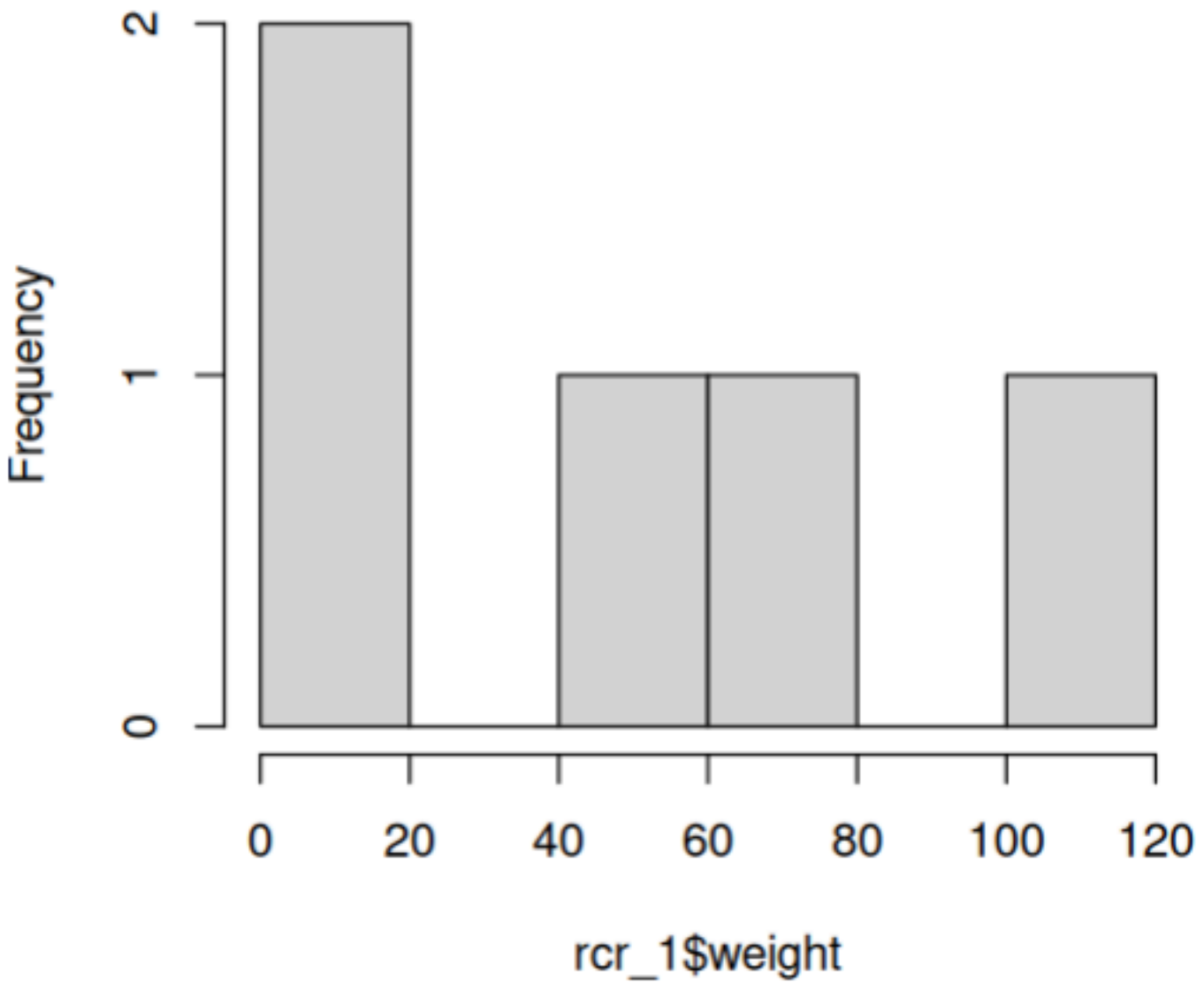


Figure 13: histogram-weight

```
summary(rca_1)
#> record_id      name_first      name_last      address
#> Length:5      Length:5      Length:5      Length:5
#> Class :character Class :character Class :character Class :character
#> Mode :character Mode :character  Mode :character Mode :character
#>
#>
#>
#> telephone      email      dob
#> Length:5      Length:5      Min. :1934-04-09 00:00:00
#> Class :character Class :character 1st Qu.:1952-11-02 00:00:00
#> Mode :character Mode :character Median :1955-04-15 00:00:00
#>                                     Mean :1969-11-05 23:48:00
#>                                     3rd Qu.:2003-03-10 00:00:00
```

```

#>                                Max.    :2003-08-30 00:00:00
#>    age                sex    demographics_complete    height    weight
#> Length:5            Female:2    Incomplete:0        Min.    : 6.0    Min.    : 1
#> Class :character    Male :3    Unverified:0        1st Qu.: 7.0    1st Qu.: 1
#> Mode  :character                Complete :5        Median :165.0    Median : 54
#>                                Mean  :110.2    Mean   : 48
#>                                3rd Qu.:180.0    3rd Qu.: 80
#>                                Max.   :193.0    Max.   :104
#>    bmi                comments                mugshot                health_complete
#> Min.    : 19.8    Length:5                Length:5                Incomplete:2
#> 1st Qu.: 24.7    Class :character    Class :character    Unverified:1
#> Median : 27.9    Mode  :character    Mode  :character    Complete :2
#> Mean   :110.9
#> 3rd Qu.:204.1
#> Max.   :277.8
#>    race__1    race__2    race__3    race__4    race__5
#> Unchecked:4    Unchecked:4    Unchecked:4    Unchecked:4    Unchecked:1
#> Checked :1    Checked :1    Checked :1    Checked :1    Checked :4
#>    race__6                ethnicity interpreter_needed
#> Unchecked:4    Unknown / Not Reported:1    Mode :logical
#> Checked :1    NOT Hispanic or Latino:3    FALSE:3
#>                Hispanic or Latino :1    TRUE :1
#>                NA's :1
#> race_and_ethnicity_complete
#> Incomplete:1
#> Unverified:0
#> Complete :4

summary(lm(age ~ 1 + sex + bmi, data = rca_1))
#>
#> Call:
#> lm(formula = age ~ 1 + sex + bmi, data = rca_1)
#>
#> Residuals:
#>      1      2      3      4      5
#> -2.491  1.954  9.132  2.491 -11.086
#> attr("label")
#> [1] "Age (years)"
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept) 63.34496    8.89980   7.118  0.0192 *
#> sexMale     13.55626    9.62958   1.408  0.2945
#> bmi        -0.24426    0.04337  -5.632  0.0301 *
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 10.55 on 2 degrees of freedom
#> Multiple R-squared:  0.9442, Adjusted R-squared:  0.8884
#> F-statistic: 16.92 on 2 and 2 DF, p-value: 0.05581

```

Note at this point there are already differences in the data, but not the fitted model parameters. The variable for sex in the REDCapR is presented as the numeric code, whereas the redcapAPI version converted it to a factor utilizing the defined metadata. How these choices of data type conversion are made and options for specifying them are covered later.

```
summary(rcr_1$sex) # REDCapR data in tibble
#>   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>   0.0     0.0     1.0     0.6     1.0     1.0
```

```
summary(rca_1$sex) # redcapAPI data in base R
#> Female  Male
#>      2     3
```

Both libraries have advantages and disadvantages when compared. The biggest consideration at this point is a user writing code to analyze data must choose a path as the downstream code working on the data is not compatible and cannot be easily swapped out. What support does one have with their department and what code are is shared for analysis is a strong consideration.

Pause for questions.

6 Be Nice to Your Server

When you read a dataset for the first time, you probably haven't decided which columns are needed so it makes sense to retrieve everything. As you gain familiarity with the data and with the analytic objectives, consider being more selective with the variables and rows transported from the remote server to your local machine.

Advantages include:

1. A server is almost always more efficient filtering than a language like R or Python.
2. REDCap's PHP code retrieves less data from REDCap's database and translates less to a text format (like csv or json).
3. Fewer bytes are transmitted across your network.
4. Your local machine will have better performance, because R has a smaller dataset to manage.
5. Your brain doesn't have to look past unnecessary columns.
6. Your R code doesn't have filter what the server already removed.
7. Highly-sensitive PHI columns that are unnecessary for an analysis remain on the server.

6.1 Specify Rows and Columns

As a basic demo of both libraries capacity to limit a query, let's make our query only pull records 1 and 4, and only the fields (columns): 'record_id', 'name_first', and 'age'.

```
records <- c(1, 4)
fields <- c("record_id", "name_first", "age")
REDCapR::redcap_read(
  redcap_uri = uri,
  token      = token,
  fields     = fields,
  records    = records,
  verbose    = FALSE
)$data
#> # A tibble: 2 × 3
#>   record_id name_first  age
#>   <dbl>    <chr>    <dbl>
#> 1         1 Nutmeg     11
#> 2         4 Trudy      61
```

```
exportRecordsTyped(
  conn,
  fields = fields,
  records = records)
#> record_id name_first age
```

```
#> 1      1      Nutmeg  11
#> 2      4      Trudy   61
```

6.1.a Row Filtering

REDCapR offers a useful filter command. It limit rows via an expression before returning. See your server's documentation for the syntax rules of the filter statements. Remember to enclose your variable names in square brackets. Also be aware of differences between strings and numbers.

```
REDCapR::redcap_read(
  redcap_uri = uri,
  token      = token,
  filter_logic = "'2003-01-01' < [dob]",
  verbose    = FALSE,
  fields     = fields
)$data
#> # A tibble: 2 × 3
#>   record_id name_first  age
#>   <dbl>    <chr>    <dbl>
#> 1       1      Nutmeg      11
#> 2       2      Tumtum      11
```

Similar limitations of a query exist for events and forms.

7 Security

When using the API as we've seen one needs an API key to get access. The API key is a user name and password in one convenient package. All the recommendations around dealing with a password apply to it's treatment and usage. This presents a problem, as writing a password into a file to get the needed access creates a high security risk.

In 2023 it is estimated that 12 million API keys were leaked on GitHub, a public code sharing website.[Toulas 2023][1]. Private health information (PHI) in the United States carries a minimum \$141 per record leaked with a cap of \$2,134,831 per incident as of August 2024[Federal Register][2]. The possibility of leakage from an API key in code is massive and should never be done. Leaking a key is a reportable security incident if PHI is involved and can have serious consequences.

BAD: Writing an API_KEY in your R code.

NOT GOOD: Putting your API_KEY in a file inside the project directory. Just one slip away from sharing, and if you're publishing your project to github that share is the world.

BETTER: Put your API_KEYS in a file outside the project directory.

8 Security Method 1: Token File (REDCapR)

The basic goals are (a) separate the secret values from the R file into a dedicated file and (b) secure the dedicated file. If using a git repository, prevent the file from being committed with an entry in .gitignore. Ask your institution's IT security team for their recommendation.

The `retrieve_credential_local()` function in the REDCapR package loads relevant information from a csv into R. The plain-text file might look like this:

```
redcap_uri,username,project_id,token,comment
"https://redcap-dev-2.ouhsc.edu/redcap/
api/", "myusername", "33", "9A068C425B1341D69E83064A2D273A70", "simple"
"https://redcap-dev-2.ouhsc.edu/redcap/
api/", "myusername", "34", "DA6F2BB23146BD5A7EA3408C1A44A556", "longitudinal"
"https://redcap-dev-2.ouhsc.edu/redcap/
api/", "myusername", "36", "F9CBFFF78C3D78F641BAE9623F6B7E6A", "simple-write"
```

To retrieve the credentials for the first project listed above, pass the value of “33” to `project_id`.

```
path_credential <- system.file("../tokenstore/dev-2.credentials", package = "REDCapR")
credential <- REDCapR::retrieve_credential_local(
  path_credential = path_credential,
  project_id      = 33
)

credential
#> $redcap_uri
#> [1] "https://redcap-dev-2.ouhsc.edu/redcap/api/"
#>
#> $username
#> [1] "myusername"
#>
#> $project_id
#> [1] 33
#>
#> $token
#> [1] "9A068C425B1341D69E83064A2D273A70"
#>
#> $comment
#> [1] "simple"
```

9 Security Method 2: Encrypted Keyring (shelter)

A plain text file with ones password on a laptop is a risk if that laptop is stolen or hacked. Storing the tokens in an encrypted manner is even better security.

GREAT: Store tokens using encryption.

Working with cryptography and security is in general more difficult than just writing the `API_KEY` into a file. The knowledge required, and coding, much less knowledge of best practices can be challenging. What’s needed is a solution that does the following:

- Follows best security practices possible.
- Usage is easier than self management of a key.
- Works in automated environments with *no code changes*.
- Works for Windows, Mac and Linux.
- Works inside RStudio and command line.

This is a tall bill of goods to accomplish, but it’s available in the R package `[shelter]`[3]. It took three years of user feedback and design to hit these goals, and this is now available with a simple interface for any `API_KEY` desired. It utilizes an encrypted key ring to store API keys to disk using encryption. Thus API keys only exist in memory unencrypted and go away when a session is closed (*assuming R isn’t saving session to disk which defeats the point of encryption!!!*).

10 shelter and redcapAPI

Let’s first look at an example using `redcapAPI`. This has a helper function (calling `shelter`) that hides a part we’ll explore later. This would be near the top of an R function to open the connection.

```
uri <- "https://redcap-dev-2.ouhsc.edu/redcap/api/"
redcapAPI::unlockREDCap(
  c(rcc = 'REDCapCon2025'),
  keyring='API_KEYS',
  envir=1,
  url=uri)
```

And with that small bit of code, the chance of leakage of `API_KEY` is greatly reduced.

The first argument `c(rcc = 'REDCapCon2025')` is a named list of connections to make. In this example it says to write to the current env `envir=1` the variable `rcc` to hold a `redcapConnection` object. The name `'REDCapCon2025'` is a user defined name that will be used in interaction. In general it's best of this is the name of the project in REDCap, but as that can change there is no constraint on the name. It is simply how the user wishes it to be known.

The second argument `keyring='API_KEYS'` says to store the requested keys in a key ring named `'API_KEYS'`. One could use the same key ring on different projects, or have just one key ring for everything. This is once again up to how the user wishes to organize their key rings. We recommend that you have a key ring for each group of related REDCap projects.

The `envir` argument just allows for writing the connections directly into the current environment as variables.

The `url` argument specifies the remote url/uri utilized for the connection.

The first time this is run, it will ask for a password for the key ring. Later executions of the command will ask for the password to unlock the key ring.

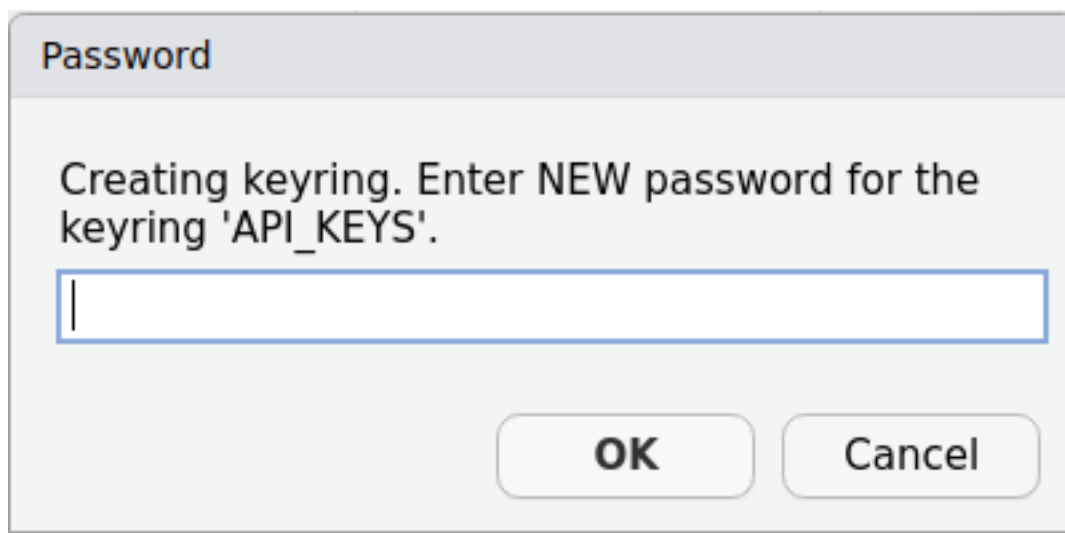


Figure 14: shelter-password

If a given API key is not found in the key ring, the user will be prompted for the named API key. Then it will test that key to see if it is valid.

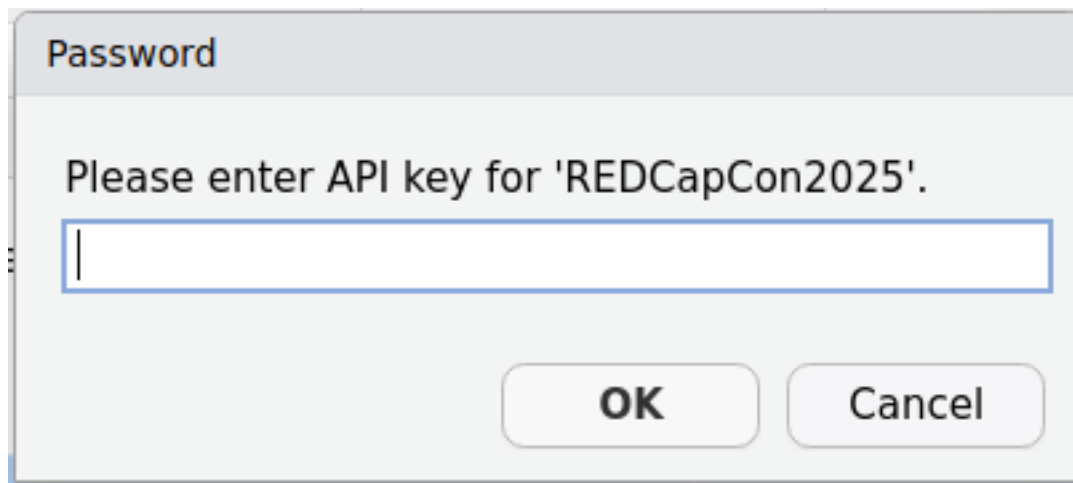


Figure 15: shelter-apikey

If validation succeeds it will return that object to the environment, if it fails it will ask again. If an API key for a project changes, it will detect this as a failure when it tries to open and ask for that key again.

11 shelter and REDCapR

REDCapR doesn't have a utility function like `redcapAPI` for using `shelter` yet (stay tuned). This gives a great opportunity to show how one can use `shelter` with any package needing API keys. I.e., `shelter` is a security package for any secret storage retrieval needs.

What's needed is a “connect and check” function. A function that given an API key, returns either a connection object or NULL upon failure. The `redcap` version is the lowest overhead on the remote server, so that call is a great check to see if an `API_KEY` is valid.

```
library(shelter)
library(REDCapR)
checkToken <- function(key, ...)
{
  ver <- redcap_version(token=key, verbose=FALSE, ...)
  if(ver == '0.0.0') NULL else key
}
```

Let's see that in action:

```
uri <- "https://redcap-dev-2.ouhsc.edu/redcap/api/"

# THIS IS EXACTLY WHAT SHOULD NEVER BE IN ONES CODE
#
token <- "9A068C425B1341D69E83064A2D273A70" # Public DEMO Project
#
# This key is to a project provided for public demo use.

bad <- paste(rep('0',32), collapse='')

checkToken(bad, redcap_uri = uri)
#> NULL
#>
checkToken(token, redcap_uri = uri)
#> [1] '15.2.0'
#>
checkToken(token, redcap_uri="https://google.com")
#> NULL
```

Thus one can use this function to store the token in a key ring instead of writing it in code via `shelter`.

```
shelter::unlockKeys(
  c(token='REDCapCon2025'),
  keyring='API_KEYS',
  connectFUN=checkToken,
  envir=1,
  redcap_uri = uri) # NOTE: ... is passed to checkToken
#>
token
#> [1] "9A068C425B1341D69E83064A2D273A70"
redcap_version(token=token, verbose=FALSE, redcap_uri = uri)
#> [1] '15.2.0'
```

Thus this token is now hidden behind the name “REDCapCon2025” and sits in memory as `token`. It is stored encrypted and is decrypted using the user supplied password. The key ring is named ‘API_KEYS’.

12 Important Note

Use of `shelter` alone is insufficient for good security. One must make sure to never save workspace to `.Rdata` on exit and never set `cache=TRUE` in `quarto` or `rmarkdown` files for a block that contains private health information (PHI) or private identifying information (PII).

In RStudio, look under Tools -> Global Options -> General and make sure save is set to NEVER and load is not clicked. This should be done for history as well.

Further making sure if one needs to store PHI/PII locally that is always stored to an encrypted volume in some form is important.

13 Security Method 3:

Create a custom database to store tokens. See writeup.

14 Takeaway on Security

Good security around sensitive data doesn't have to be difficult, and now one is equipped with practical and easy means to prevent key leakage. Utilizing `shelter::unlockKeys` or `redcapAPI::unlockREDCap` has an additional advantage that it works well with automated environments creating a seamless transition to an automated report system. A sys admin may utilize overriding the interactive bits of `shelter` by providing system level ENV variables or a local yml key file. Thus with a simple call one is now enhancing security *and* preparing for automated reporting.

If you have automation needs have your sys admin email on of the authors of the `shelter` package. Details will be provided.

14.1 References

- [1]: Toulas B (2023). "Over 12 million auth secrets and keys leaked on GitHub in 2023." <https://www.bleepingcomputer.com/news/security/over-12-million-auth-secrets-and-keys-leaked-on-github-in-2023/>
- [2]: (2024). Federal Register, (2024-17446), 6481501364832.
- [3]: CRAN: shelter

15 Type Theory (type casting)

One of the major benefits of utilizing a pre-existing package like `REDCapR` or `redcapAPI` is the amount of reuseable code that has gone into it's design. In the heart of these packages something known as 'type casting' occurs and it is a central concept which contains a large amount of value added.

Type casting is the conversion from one data type into another in a computer. All data coming from `REDCap` is a character based string, and R has a variety of data types utilized in analysis, e.g. numeric or factor. A naïve idea would be that when given a string "123" it going to be easy to convert into a numeric in R. For that specific string it is, but there are a huge number of concerns that lurk in those waters. We shall dive into a short aside on just how theoretically deep those waters are and a follow up of the practicalities of `REDCap` data.

The following paragraph can be safely ignored.

$f : A \rightarrow B$ denotes a function, f in type theory is a function that converts something of type A into something of type B . The value of automatically evaluating and checking this has become very important in computer science. However, the idea has become incredibly important in mathematics as well. Bertrand Russell published *Principia Mathematica* in 1910 seeking to find a single foundation for all of mathematics. He settled on set theory and it had 18 axioms, small formal assumptions, which it was based on. In this system it takes about 300 pages of math to show that $1+1=2$. Godel's famous incompleteness theorem in 1929 was a huge blow to the goals of finding a single system that all mathematics could be based upon—as any such system must forever remain incomplete. Type theory in computers began making theoretical advances in the 1960's, a system Russell had initially considered. Voevodsky in 2006 showed a homotopy between λ -calculus, algebraic topology and type theory which is homotopy type theory (HoTT). In this foundation only 2 axioms are required to bootstrap mathematics and a short few pages to show $1+1=2$. Russell's vision from a century before has been realized. One of those axioms, is the existence of a path between two points A and B which is represented by $f : A \rightarrow B$, thus making type casting one of the deepest philosophical entities in mathematics!

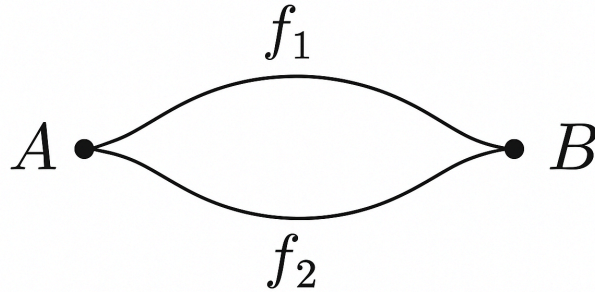


Figure 16: type-theory

Fortunately, a REDCap user's needs are simpler—and the API developers have you covered. However, it is not as simple as it sounds. All data stored in REDCap is character or string. In the computer, an analyst wants dates, numbers, factors or potentially some other data type the authors of REDCapR and redcapAPI have never heard of. To further complicate matters data can be missing under a variety of definitions and it might be invalid as user input may have been changed from free form to date specified in the middle of data collection leading to a huge number of uninterpretable date values.

Type casting from REDCap concerns itself with 3 fundamental steps:

1. Is a value NA?
2. Is a value valid given it's specification in the REDCap data dictionary?
3. For values that are not NA and are valid, convert in memory to the computer's related type.

Secondly, the inverse applies when writing data back via the API. Given a value in memory, type cast it down to a string that can be stored properly in REDCap.

Both libraries both make choices for the user that are the usual for 95% of the cases, but that doesn't mean those choices are what ones project needs. Thus inversion of control is provided such that the user can override these operations with their own.

15.1 REDCapR and readr Approach

As the automation of your scripts matures and institutional resources depend on its output, its output should be stable. One way to make it more predictable is to specify the column names and the column data types. In the previous example, notice that R (specifically `readr::read_csv()`) made its best guess and reported it in the "Column specification" section.

In the following example, REDCapR passes `col_types` to `readr::read_csv()` as it converts the plain-text output returned from REDCap into an R data frame. (To be precise, a tibble is returned.)

When `readr` sees a column with values like 1, 2, 3, and 4, it will make the reasonable guess that the column should be a double precision floating-point data type. However we recommend using the simplest data type reasonable because a simpler data type is less likely contain unintended values and it's typically faster, consumes less memory, and translates more cleanly across platforms. As specifically for identifiers like `record_id` specify either an integer or character.

```
# Specify the column types.
desired_fields <- c("record_id", "race")
col_types <- readr::cols(
  record_id = readr::col_integer(),
  race__1   = readr::col_logical(),
  race__2   = readr::col_logical(),
  race__3   = readr::col_logical(),
```

```

race___4 = readr::col_logical(),
race___5 = readr::col_logical(),
race___6 = readr::col_logical()
)
REDCapR::redcap_read(
  redcap_uri = uri,
  token      = token,
  fields     = desired_fields,
  verbose    = FALSE,
  col_types  = col_types
)$data
#> # A tibble: 5 x 7
#>   record_id race___1 race___2 race___3 race___4 race___5 race___6
#>   <int> <lgl> <lgl> <lgl> <lgl> <lgl> <lgl>
#> 1     1     FALSE     FALSE     FALSE     FALSE     TRUE     FALSE
#> 2     2     FALSE     FALSE     TRUE      FALSE     TRUE     FALSE
#> 3     3     FALSE     FALSE     FALSE     TRUE      TRUE     FALSE
#> 4     4     FALSE     TRUE      FALSE     FALSE     TRUE     FALSE
#> 5     5      TRUE     FALSE     FALSE     FALSE     FALSE     TRUE

```

15.1.a REDCapR Specify Everything is a Character

REDCap internally stores every value as a string. To accept full responsibility of the data types, tell `readr::cols()` to keep them as strings.

```

# Specify the column types.
desired_fields <- c("record_id", "race")
col_types <- readr::cols(.default = readr::col_character())
REDCapR::redcap_read(
  redcap_uri = credential$redcap_uri,
  token      = credential$token,
  fields     = desired_fields,
  verbose    = FALSE,
  col_types  = col_types
)$data
#> # A tibble: 5 x 7
#>   record_id race___1 race___2 race___3 race___4 race___5 race___6
#>   <chr> <chr> <chr> <chr> <chr> <chr> <chr>
#> 1 1     0     0     0     0     1     0
#> 2 2     0     0     1     0     1     0
#> 3 3     0     0     0     1     1     0
#> 4 4     0     1     0     0     1     0
#> 5 5     1     0     0     0     0     1

```

15.2 redcapAPI and specifying conversion

The package `redcapAPI` provides 3 sets of callback lists: `na`, `validation` and `cast` to split the process into much finer granular pieces. This allows far deeper control over the conversion steps by the user, but comes at a cost of greater complexity. The default conversion steps are the most widely used as observed by the authors.

Let's dive in with an example of the normal output.

```

redcapAPI::exportRecordsTyped(
  conn,
  fields = desired_fields
)
#>   record_id race___1 race___2 race___3 race___4 race___5 race___6
#> 1         1  Unchecked Unchecked Unchecked Unchecked   Checked Unchecked

```

```
#> 2      2 Unchecked Unchecked   Checked Unchecked   Checked Unchecked
#> 3      3 Unchecked Unchecked Unchecked   Checked   Checked Unchecked
#> 4      4 Unchecked   Checked Unchecked Unchecked   Checked Unchecked
#> 5      5   Checked Unchecked Unchecked Unchecked Unchecked   Checked
```

The default choice the library made was to make the checkboxes an R factor and label them as ‘Checked’ or ‘Unchecked’. Let’s let the user make a choice and specify logical.

```
redcapAPI::exportRecordsTyped(
  conn,
  fields = desired_fields,
  cast   = list(checkbox=castLogical)
)
#>   record_id race__1 race__2 race__3 race__4 race__5 race__6
#> 1         1   FALSE   FALSE   FALSE   FALSE   TRUE   FALSE
#> 2         2   FALSE   FALSE   TRUE   FALSE   TRUE   FALSE
#> 3         3   FALSE   FALSE   FALSE   TRUE   TRUE   FALSE
#> 4         4   FALSE   TRUE   FALSE   FALSE   TRUE   FALSE
#> 5         5    TRUE   FALSE   FALSE   FALSE   FALSE   TRUE
```

The viewpoint from readr was column (field) based, but here it’s field description based. One can write custom functions that switch based on name if desired. It is possible to do the same functionality as readr but requires more coding.

15.3 redcapAPI everything character

If one desired the absolute raw character values stored, with redcapAPI there is an additional need to override validations— as these values will get dropped by the engine.

```
redcapAPI::exportRecordsTyped(
  conn,
  fields      = desired_fields,
  validation  = skip_validation,
  cast        = raw_cast
)
#>   record_id race__1 race__2 race__3 race__4 race__5 race__6
#> 1         1      0      0      0      0      1      0
#> 2         2      0      0      1      0      1      0
#> 3         3      0      0      0      1      1      0
#> 4         4      0      1      0      0      1      0
#> 5         5      1      0      0      0      0      1
```

One thing that is quite common as an analyst is locating those pesky values that do not cast properly. This is payoff from having validation as a separate step.

```
> reviewInvalidRecords(redcapAPI::exportRecordsTyped(conn))
#> # Failed Validations from REDCap project 'REDCapR: simple'
#>
#> Tue 09 Sep 2025 12:00:00 AM UTC
#> Package redcapAPI version 2.11.2
#> REDCap version 15.2.0
```

There’s no data validations to report, all is good. The print for this renders to markdown for embedding in a report. If there had been validation failures—it would provide a direct html link to the form to edit that record.

16 Zero Coding

A rare issue that has occurred in practice is that of a variable that has zero coded value. Typically 0 indicates lack of data or NA, but 0 may have a meaning in some contexts. An extreme example is where if one were to define 0=TRUE and 1=FALSE, there are cases where type casting is indeterminate. redcapAPI issues a warning if it encounters this case to encourage the user to spot audit data that has zero coding to ensure things are as expected and the argument `warn_zero_coding=FALSE` will turn off this message.

17 Sparse Block Matrix Form

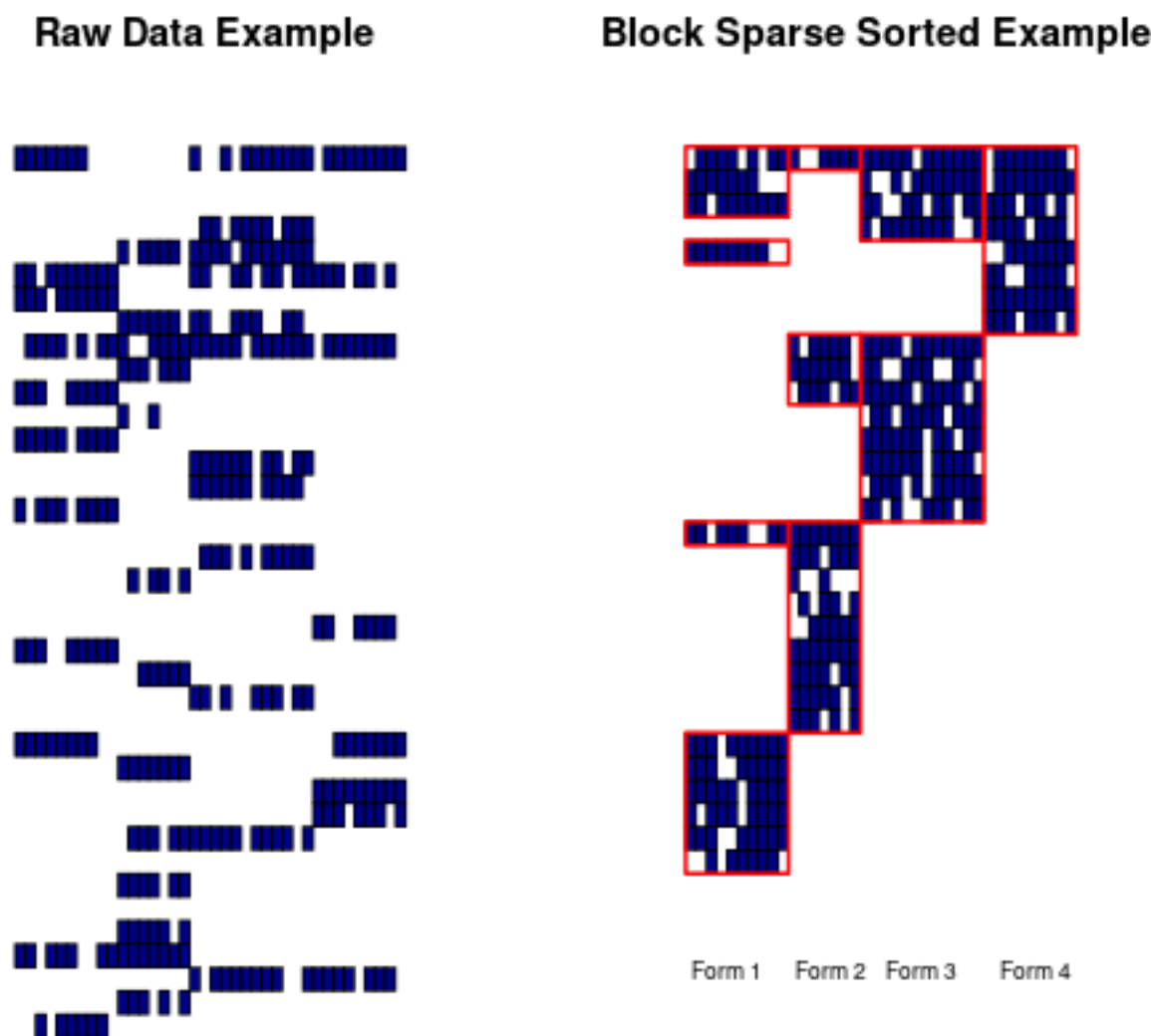


Figure 17: block_sparse

Another issue in play is the context in which exporting data from REDCap without restriction to a form means that every column for every form has a value. This results in a block sparse format with the majority of the data being NULL and potentially thousands of columns wide. It's best to filter and extract these down to the relevant forms.

17.1 redcapAPI export forms

One method of dealing with this is a wrapper function in `redcapAPI::exportBulkRecords`. This is a loop function that queries the meta data, finds all the defined forms, and then requests the data for each form with only the appropriate columns. Empty rows are dropped. This can greatly reduce server and network load while reducing time to produce analytically ready data.

```
conn$instruments()
#>      instrument_name  instrument_label
#> 1      demographics      demographics
#> 2           health           health
#> 3 race_and_ethnicity race_and_ethnicity

exportBulkRecords(list(simple=conn))$simple_demographics
#>   record_id name_first name_last address
#> 1         1   Nutmeg   Nutmouse 14 Rose Cottage St.\nKenning UK, 323232
#> 2         2   Tuntum   Nutmouse 14 Rose Cottage Blvd.\nKenning UK 34243
#> 3         3   Marcus   Wood      243 Hill St.\nGuthrie OK 73402
#> 4         4    Trudy    DAG       342 Elm\nDuncanville TX, 75116
#> 5         5  John Lee   Walker    Hotel Suite\nNew Orleans LA, 70115
#>   telephone      email      dob age  sex demographics_complete
#> 1 (405) 321-1111 nutty@mouse.com 2003-08-30 11 Female Complete
#> 2 (405) 321-2222 tummy@mouse.comm 2003-03-10 11 Male Complete
#> 3 (405) 321-3333 mw@mwood.net 1934-04-09 80 Male Complete
#> 4 (405) 321-4444 peroxide@blonde.com 1952-11-02 61 Female Complete
#> 5 (405) 321-5555 left@hippocket.com 1955-04-15 59 Male Complete
```

Coupled with `unlockKeys`'s ability to open multiple project, `exportBulkRecords` can loop over and pull all the data in forms directly into memory making ready to analyze data organized and type cast in two carefully constructed calls!

18 Checkboxes

With so many concerns handled in a consistent, repeatable manner by the libraries there still remains one difficult issue—that of the checkbox.

The checkbox from a user experience using a form is a wonderful thing. A user can click/unclick a box and move on quickly. The root of the problem is that a checkbox *always* has a value—checked or unchecked. There is no possibility of an NA by *definition*. Just by including a checkbox on a form—it now always has a value on that form. Couple this with the issue of detecting forms via sparse block matrices and whether a checkbox is NA is now dependent on the context in which it exists and it is not uncommon to define it as NA when all the other fields in the same form are NA or unchecked if a checkbox. This presents contextual interpretation of a checkbox's value which is not a desirable thing.

Checkboxes are best avoided and a dropdown that has a yes/no in which a user explicitly enters a value is a preferable alternative. Thus a user has explicitly set the value and the interpretation is clear.

Be aware that checkboxes will cause difficulties in getting data clean for analysis and custom filtering code to remove empty rows based on an analysts interpretation is required.

19 Longitudinal

19.1 Background

This section of the presentation pertains to reading REDCap records from a project that (a) has longitudinal events or (b) has a repeating measure. The first section conceptually discusses how REDCap stores complex structures. The remaining sections describe how to best retrieve complex structures with the REDCapTidyieR and REDCapR packages.

- If you are new to R or REDCap, consider start with the Typical REDCap Workflow for a Data Analyst and Basic REDCapR Operations vignettes and then return to this document.
- If you are reading from a *simple* project, just call REDCapR's `redcap_read()`.

- If you want to perform some other operation (such as writing records to REDCap), review the Reference of REDCapR functions to see what is currently available.

If your REDCap project is longitudinal or contains repeating measures, a single call to the API (or a single export through the browser) will return a dataset that is not readily analyzed. Instead, the dataset will resemble Table 5. This isn't because of a software bug, but because you haven't told the software how you would like the data structured.

There isn't a good way to jam this multidimensional space into a rectangle of points. Our advice for querying REDCap is the same as querying any database system: request separate datasets that have a natural "grain" and assemble them as to fit your analyses.

19.2 Illustration of How Data Points are Structured

19.2.a Possible Table Structures

Suppose you have two patients (*i.e.*, "1" and "2") with three intake variables (*i.e.*, height, weight, and bmi). If you record this on a piece of paper, it would probably look like Table 1. The table's *grain* is "patient", because each row represents a distinct patient. Understanding the grain of each structure below will help you understand how the structures are re-expressions of the same set of observations.

19.2.a.a Table 1: patient grain

ptid	height	weight	bmi
1	1.0	11.0	111.0
2	2.0	22.0	222.0

This patient-grain structure is how the data points are most comfortably inputted by humans into REDCap, and it is the default when exported through the browser and API. However it is stored differently by REDCap's internal database.

REDCap's flexibility is a driver of its success. Once a research team learns REDCap, it can reuse the knowledge to capture anything from leukemia to lunch orders. But to achieve this flexibility in the world of REDCap and EMRs, data are stored along the observation grain. In computer science, this is commonly called an EAV structure (which stands for entity-attribute-value). The patient's ID is the entity, the variable type is the attribute, and the observed point is the value. It can also be thought of as a "key-value store" nested within a patient (where "key" is a synonym of "attribute"). Notice that the two wider rows have morphed into six skinnier rows –one row per observation. If you are a curious database administrator, peek at the the structure and rows of the `redcap_data` table. It is the most important table in the database.

19.2.a.b Table 2: observation grain for intake instrument

REDCap and EMR databases store observations in their underlying table. This table is a simplification of the `redcap_data` table, which is the heart of the REDCap's internal database.

ptid	key	value
1	height	1.0
1	weight	11.0
1	bmi	111.0
2	height	2.0
2	weight	22.0
2	bmi	222.0

If the investigation gains a longitudinal or repeating component, it becomes necessary to include the dimension of time. Suppose the protocol specifies five time points; the blood pressure instrument is captured at times 1, 2, & 3 while the laboratory instrument is captured at times 1 & 2. If you record this stage on paper, it will likely resemble Tables 3a & 3b: one for vitals and one for labs.

19.2.a.c Table 3a: patient-time grain for blood_pressure instrument

ptid	time	sbp	dbp
1	1	1.1	11.1
1	2	1.2	11.2
1	3	1.3	11.3
2	1	2.1	22.1
2	2	2.2	22.2
2	3	2.3	22.3

19.2.a.d Table 3b: patient-time grain for laboratory instrument

ptid	time	lab	dose
1	1	aa1	1.1 mg
1	2	aa2	1.2 mg
2	1	bb1	2.1 mg
2	2	bb2	2.2 mg

When these measurements are added to REDCap's observation table, it resembles Table 4. Two new columns are required to uniquely distinguish the instrument and its ordinal position. Notice the first six rows are copied from Table 2; they have empty values for the repeating structure.

19.2.a.e Table 4: observation grain for intake, blood_pressure, and laboratory instruments

ptid	repeatinstrument	repeatinstance	key	value
1	–	–	height	1.0
1	–	–	weight	11.0
1	–	–	bmi	111.0
2	–	–	height	2.0
2	–	–	weight	22.0
2	–	–	bmi	222.0
1	blood_pressure	1	sbp	1.1
1	blood_pressure	1	dbp	11.1
1	blood_pressure	2	sbp	1.2
1	blood_pressure	2	dbp	11.2
1	blood_pressure	3	sbp	1.3
1	blood_pressure	3	dbp	11.3
1	laboratory	1	lab	aa1
1	laboratory	1	conc	1.1 ppm
1	laboratory	2	lab	aa2
1	laboratory	2	conc	1.2 ppm
2	blood_pressure	1	sbp	2.1
2	blood_pressure	1	dbp	22.1

ptid	repeatinstrument	repeatinstance	key	value
2	blood_pressure	2	sbp	2.2
2	blood_pressure	2	dbp	22.2
2	blood_pressure	3	sbp	2.3
2	blood_pressure	3	dbp	22.3
2	laboratory	1	lab	bb1
2	laboratory	1	conc	2.1 ppm
2	laboratory	2	lab	bb2
2	laboratory	2	conc	2.2 ppm

As mentioned above, there isn't a universally good way to coerce Tables 1, 3a, and 3b into a single rectangle because the rows represent different things. Or from REDCap's perspective, there's not a good transformation of redcap_data (*i.e.*, Table 4) that is appropriate for most statistical programs.

When forced to combine the different entities, the best option is probably Table 5. We call this a "block dataset", borrowing from linear algebra's block matrix term. You can see the mishmash of tables masquerading as a unified dataset. The rows lack the conceptual coherency of Tables 1, 3a, & 3b.

19.2.a.f Table 5: mishmashed grain

ptid	repeatinstrument	repeatinstance	height	weight	bmi	sbp	dbp	lab	conc
1	–	–	1.0	11.0	111.0	–	–	–	–
1	blood_pressure	1	–	–	–	1.1	11.1	–	–
1	blood_pressure	2	–	–	–	1.2	11.2	–	–
1	blood_pressure	3	–	–	–	1.3	11.3	–	–
1	laboratory	1	–	–	–	–	–	aa1	1.1 ppm
1	laboratory	2	–	–	–	–	–	aa2	1.2 ppm
2	–	–	2.0	22.0	222.0	–	–	–	–
2	blood_pressure	1	–	–	–	2.1	22.1	–	–
2	blood_pressure	2	–	–	–	2.2	22.2	–	–
2	blood_pressure	3	–	–	–	2.3	22.3	–	–
2	laboratory	1	–	–	–	–	–	bb1	2.1 ppm
2	laboratory	2	–	–	–	–	–	bb2	2.2 ppm

A block dataset is not inherently bad. After all, Table 5 can be unambiguously transformed to and from Table 4.

Table 5's primary limitation is that a block dataset is not understood by analysis software used in conventional medical research. At best, the dataset always will require additional preparation. At worst, the analyst will model the rows inappropriately, which will produce misleading conclusions.

Table 5's secondary limitation is inefficiency. The empty cells aren't computationally free. Every cell must be queried from the database and concatenated in REDCap's web server in order to return Table 5 in the plain-text csv, json, or xml format. In our simple example, more than half of the block dataset's cells are wasted. The emptiness frequently exceeds 90% in real-world REDCap projects (because they tend to have many more variables and repeating instances). The emptiness always exceeds 99.9% in real-world EMRs.

For this reason, REDCap and EMR design their observation table to resemble the computational structure of a sparse matrix. (The only important difference is that REDCap’s unspecified cells are interpreted as null/empty, while a sparse matrix’s unspecified cells are interpreted as zero.)

In the case of a sparse matrix, substantial memory requirement reductions can be realized by storing only the non-zero entries. Depending on the number and distribution of the non-zero entries, different data structures can be used and yield huge savings in memory when compared to the basic approach. The trade-off is that accessing the individual elements becomes more complex and additional structures are needed to be able to recover the original matrix unambiguously. (source: Wikipedia: Sparse matrix - storage)

19.2.b Terminology

19.2.b.a observation

The term “observation” in the world of medical databases has a different and more granular meaning than it does in the tidyverse literature. In REDCap and medical databases, an observation is typically a single point (such as a heart rate or systolic blood pressure) with contextual variables (such as the the associated date, unit, visit ID, and patient ID); see Tables 2 and 4 above. In the tidyverse publications, an observation is roughly equivalent to a REDCap instrument (which is a collection of associated values); see Tables 1, 3a, and 3b.

Concept	REDCap & Medical World	Tidyverse Literature
A single measured point	observation	value
A collection of associated points	instrument	observation

19.3 Retrieving from REDCap

Many new REDCap users will submit a single API call and unintentionally obtain something like Table 5; they then try to extract something resembling Tables 1, 3a, & 3b. Although this can be successful, we strongly discourage it. The code is difficult to maintain and is not portable to REDCap projects with different instruments. (The code is really slow and ugly too.)

Our advice is to start before Table 5 is assembled –retrieve the information in a better way. Like other database systems, request the three tables separately from the server and then combine them on your desktop to fit your analyses if necessary.

Two approaches are appropriate for most scenarios:

1. multiple calls to REDCapR’s `redcap_read()`, or
2. a single call to REDCapTidieR’s `redcap_read_tidy()`.

Today’s presentation uses these credentials to retrieve the practice/fake dataset. **This is not appropriate for datasets containing PHI or other sensitive information.** Please see Part 2 - Retrieve Protected Token of the Typical REDCap Workflow for a Data Analyst vignette for secure approaches.

Or even better, use something like Shawn’s “shelter” package w/ PHI. For more context, Univ of Oklahoma uses a database as a foundation of a token server. As Shawn said, the token storage and retrieval is independent of the package (and almost of the programming language).

```
# Retrieve token, or even better use something like the shelter package for PHI
path_credential <- system.file("misc/dev-2.credentials", package = "REDCapR")
credential <- REDCapR::retrieve_credential_local(
  path_credential = path_credential,
  project_id      = 62
)
```

19.3.a One REDCapR Call for Each Table

The tidy datasets represented in Tables 1, 3a, and 3b can be obtained by calling REDCapR three times –one call per table. Using the `forms` parameter, pass “intake” to get Table 1, “blood_pressure” to get Table 3a, and “laboratory” to get Table 3b.

Although it is not required, we recommend specifying a `readr::cols()` object to ensure the desired variable data types.

19.3.a.a Retrieve patient-level table (corresponding to Table 1)

```
col_types_intake <-  
  readr::cols_only(  
    record_id           = readr::col_integer(),  
    height              = readr::col_double(),  
    weight              = readr::col_double(),  
    bmi                 = readr::col_double()  
  )  
  
ds_intake <-  
  REDCapR::redcap_read(  
    redcap_uri = credential$redcap_uri, # From the previous code snippet.  
    token      = credential$token,  
    forms      = c("intake"),  
    col_types  = col_types_intake,  
    verbose    = FALSE,  
  )$data  
  
ds_intake
```

19.3.a.b Retrieve patient-time-level tables (corresponding to Tables 3a & 3b)

```
col_types_blood_pressure <-  
  readr::cols(  
    record_id              = readr::col_integer(),  
    redcap_repeat_instrument = readr::col_character(),  
    redcap_repeat_instance  = readr::col_integer(),  
    sbp                    = readr::col_double(),  
    dbp                    = readr::col_double(),  
    blood_pressure_complete = readr::col_integer()  
  )  
  
ds_blood_pressure <-  
  REDCapR::redcap_read(  
    redcap_uri = credential$redcap_uri,  
    token      = credential$token,  
    forms      = c("blood_pressure"),  
    col_types  = col_types_blood_pressure,  
    verbose    = FALSE,  
  )$data  
  
ds_blood_pressure |>  
  tidyr::drop_na(redcap_repeat_instrument)  
  
col_types_laboratory <-  
  readr::cols(  
    record_id              = readr::col_integer(),  
    redcap_repeat_instrument = readr::col_character(),  
    redcap_repeat_instance  = readr::col_integer(),  
    lab                    = readr::col_character(),  
    conc                   = readr::col_character(),  
    laboratory_complete     = readr::col_integer()  
  )  
  
ds_laboratory <-  
  REDCapR::redcap_read(  
    redcap_uri = credential$redcap_uri,  
    token      = credential$token,  
    forms      = c("laboratory"),  
    col_types  = col_types_laboratory,  
    verbose    = FALSE,  
  )$data
```

```

redcap_uri = credential$redcap_uri,
token      = credential$token,
forms      = c("laboratory"),
col_types  = col_types_laboratory,
verbose    = FALSE
)$data

ds_laboratory |>
  tidyr::drop_na(redcap_repeat_instrument)

```

19.3.a.c Retrieve block tables (corresponding to Table 5)

If for some reason you need the block dataset through the API, one call will retrieve it.

```

ds_block <-
  REDCapR::redcap_read(
    redcap_uri = credential$redcap_uri,
    token      = credential$token,
    col_types  = readr::cols(.default = readr::col_character()),
    verbose    = FALSE,
  )$data

ds_block

```

19.3.b One REDCapTidieR Call for All Tables

REDCapTidieR's initial motivation is to facilitate longitudinal analyses and promote tidy data hygiene.

One call to a REDCap project will return a supertibble that contains Tables 1, 3a, and 3b as separate tibbles within a list-column.

19.3.c Choosing between the Approaches

When retrieving data from REDCap, we recommend calling REDCapTidieR in many scenarios, such as:

- you are new to managing or analyzing data with R, or
- your analyses will require most of the dataset's rows or columns, or
- you'd benefit from some of the auxiliary information in REDCapTidieR's supertibble, such as the instrument's structure.

However we recommend calling REDCapR in other scenarios. It could be worth calling REDCapR multiple times if:

- you are performing some operation other than retrieving/reading data from REDCap,
- you are comfortable with managing and analyzing data with R, or
- your analyses require only a fraction of the data (such as (a) you need only the first event, or (b) the analyses don't involve most of the instruments), or
- you want to specify the variables' data types with `readr::cols()`.

If in doubt, start with REDCapTidieR. Escalate to REDCapR if your download time is too long and might be decreased by reducing the information retrieved from the server and transported across the network.

And of course many scenarios are solved best with a combination of both packages, such as (a) REDCapR populates the initial demographics in REDCap, (b) research staff enter measures collected from patients over time, (c) REDCapTidieR retrieves the complete longitudinal dataset, (d) dplyr joins the tibbles, and finally (e) lme4 tests hypotheses involving patient trajectories over time.

19.3.d Escalating to REDCapR

Even if you think you'll need REDCapR's low-level control, consider starting with REDCapTidieR anyway. ...particularly if you are unsure how to specify the grain of each table. The structure of REDCapTidieR's tables easily compatible with conventional analyses. If you need the performance of REDCapR but are unsure how the tables should look, simply execute

something like `REDCapTidieR::redcap_read_tidy(url, project_token)` and study its output. Then try to mimic it exactly with `REDCapR::redcap_read()` calls.

Finally, cull unwanted cells using the parameters of `REDCapR::redcap_read()`. These data points will not even leave the REDCap instance, which will improve performance. Some possible strategies include passing arguments to

- **forms:** this will retrieve only the specified instruments/forms. Beginners should start here. It is easy to conceptualize and usually has a big increase in speed for just a little development work.
- **events:** this will retrieve only the desired events within a longitudinal project. For instance if the analyses involve only the “intake” and “follow up #1” events, leave follow ups #2, #3, and #4 on the server.
- **fields:** this is more granular than forms. It can be combined with calls to forms, such as passing “bmi” to fields and `c("blood_pressure", "laboratory")` to forms.
- **records:** in the example scenario, this will pluck individual patients and their associated events and repeating instances. To be useful in research, it’s usually combined with other REDCapR calls. See the Read a subset of records, conditioned on the values in some variables section of REDCapR’s Basic REDCapR Operations vignette.
- **filter_logic:** this will leverage the observation values to limit the rows returned. Because the underlying table does not index the obs values, this will be less computationally efficient than the options above.
- **datetime_range_begin & datetime_range_end:** this will return only the records that have been created or modified during the specified window.

Note that the efficiency gain from moving from the block dataset to REDCapTidieR is different than the gain from moving from REDCapTidieR to REDCapR. When moving from from Table 5 to a REDCapTidieR Supertibble, you are eliminating empty cells that will never contain worthwhile data. When moving from a REDCapTidieR Supertibble call to a collection of REDCapR calls, you are eliminating cells that contain data, but may not be relevant to your analysis (such as a patient’s name or the time a lab specimen was collected).

19.3.e Transforming Long to Wide

Sometimes the data structure is long, and the analysis software/function wants wide. If so, either transform yourself with something like `tidyr::pivot_wider()`, or let `redcapAPI` leverage its knowledge of REDCap structures with `redcapAPI::widerRepeated()`.

20 Writing

20.1 Writing to the server is more error prone than reading

Mentality: start small & simple

Test the water:

- Start w/ a single patient & a few columns (eg, `record_id` & `gender`)
- Slowly add more rows & columns

Or download the dataset, and try to reupload it. Then make a little change and try to reupload it.

And intermediate snippet might be:

```
ds_patient |>
  dplyr::select(                # First three columns
    id_code,
    date,
    is_mobile,
  ) |>
  dplyr::slice(1:2) |>         # First two rows
  REDCapR::redcap_write(
    ds_to_write = _,
    redcap_uri  = credential$redcap_uri,
    token       = credential$token,
```

```
convert_logical_to_integer = TRUE
)
```

Strategy:

- Upload the patient-level instrument(s)
- Upload the each repeating instrument separately.

20.2 Gotchas

- Client encoding & problematic non-ASCII characters
- Recode Variables where necessary
- Plumbing Variables (especially for repeating structures)

```
ds_daily <-
  ds_daily |>
  dplyr::group_by(id_code) |>
  dplyr::mutate(
    redcap_repeat_instrument = "daily",
    redcap_repeat_instance   = dplyr::row_number(da_date),
    daily_complete           = REDCapR::constant("form_complete"),
  ) |>
  dplyr::ungroup() |>
  dplyr::select(
    id_code,                                     # Or `record_id`, if you didn't rename it
    # redcap_event_name,                       # If the project is longitudinal or has arms
    redcap_repeat_instrument,                  # The name of the repeating instrument/form
    redcap_repeat_instance,                    # The sequence of the repeating instrument
    tidyselect::everything(),                 # All columns not explicitly passed to `dplyr::select()`
    daily_complete,                           # Indicates incomplete, unverified, or complete
  )
```

- Utilize Helper functions, eg

```
# Check for potential problems. (Remember zero rows are good.)
REDCapR::validate_for_write(ds_daily, convert_logical_to_integer = TRUE)
```

20.3 Next Steps

- More Complexity
 - eg, arms, longitudinal events
- Batching
- Manual vs API

If you have trouble uploading, consider adding a few fake patients & measurements and then download the csv. It might reveal something you didn't anticipate. But be aware that it will be in the block matrix format (i.e., everything jammed into one rectangle.)

- REDCap's CDIS & Other interoperability
 - See Stephany Duda et al's talk in 15 min: "REDCap + NIH: A Recipe for Innovation"
 - See Adam's Birds of a Feather session 3:50pm today: "REDCap SHARE (Sharing Health Access for Research & Empowerment)"
 - See Alex, Francesco, & Adam's talk 8:50am tomorrow: "EHR Integration: What's New & What's Next"

The Clinical Data Interoperability Services (CDIS) use FHIR to move data from your institution's EMR/EHR (eg, Epic, Cerner) to REDCap. Research staff have control over which patient records are selected or eligible. Conceptually it's similar to writing to REDCap's with the API, but at much bigger scale. Realistically, it takes months to get through your institution's human layers. Once established, a project would be populated with EMR data in much less development time –assuming the desired data models corresponds with FHIR endpoints.

20.4 Resources for Writing/Importing

- Language Agnostic Troubleshooter: <https://ouhscbbmc.github.io/REDCapR/articles/TroubleshootingApiCalls.html#writing>
- REDCapR Writing Vignette: <https://ouhscbbmc.github.io/REDCapR/articles/workflow-write.html>

21 REDCap API 2.0

{Insert from <snippets/redcap-api-2.md>}

22 Further Resources

22.1 Community

- <https://redcap.vumc.org/community>
- Community is for REDCap admins, which many biostatisticians are not

22.2 REDCap Tools

- <https://redcap-tools.github.io/projects/>
- Libraries: discussed today
- Common resources: eg, Language-agnostic API Troubleshooter & FAQ
- Code samples eg, Paul Litwin's trigger code for C#
- Applications: eg, Philip Chase's REDCap Custodian

22.3 General Sites

- Stack Overflow: redcap tag
179 questions, with a lot of responses from Aidan Wilson & Luke Stevens
- Reddit: ProjectREDCap subreddit
- CRAN: Web Technologies Task View
(I've put it under "Survey, Questionnaire, and Data Capture Tools")

22.4 Your Local REDCap Admins

- Can help you
 - if they know, or
 - post a question to Community, or
 - get you in touch with a friend at another institution.
- REDCap API Playground (see Günther's slides)

22.5 Packages in Other Languages

Actively developed packages & their maintainers (who are here that you can talk to):

- Python: PyCap (Paul Wildenhain, CHOP)
 - Initially started by Scott Burns while at Vanderbilt
- PHP: PHPCap (Jim Mullen, U Indiana)
- C#: redcap-api (Michael Tran, VCU)

22.6 Other Packages in R

- REDCapTidieR (Stephan Kadauke & Richard Hanna, CHOP)
 - Focuses on a tidy *grain*
- tidyREDCap (Raymond Balise, U Miami) tomorrow 2pm biostat track
 - Focuses on a cleaner *variables*
- REDCapDM (João Carmezim, IGTP, Spain)
- REDCapCAST (Andreas Gammelgaard, Aarhus U, Denmark)
- Some other new packages that are new & exciting, but I like to wait ~2 years to see if a package remains actively maintained.

23 Closing

23.1 API & Human Connections

- We've spent an hour on the basics of REDCap connections.
- Please reach out to others at meals & after the conference

23.2 RAISE

- "REDCap and R for Increasing Scientific Efficiency"
- To join the monthly meeting: https://redcap.link/RAISE_Membership
- or email Chris Battiston with question: chris.battiston@wchospital.ca
- <https://redcap.vumc.org/community/post.php?id=260867>

23.3 Security Committee

- Committee that develops "Best Practices for Token Management"
- Goal: reach a consensus on acceptable approaches and include our recommendations in a few places like the API documentation, Community, and package/library documentation.
- Helps new API sties get started
- Helps established API sites gain trust from their org's security team.
- Helps each other stay up-to-date w/ best practices & packages
- <https://redcap.vumc.org/community/post.php?id=117300>

23.4 Contribute to Existing Packages

- Both the foundation packages (redcapAPI & REDCapR), as well as the 2nd layer packages
- Add vignettes, documentation, functions, parameters, & test cases
- Find most packages at REDCap Tools
- There's no need to compete to be recognized. It's not zero sum. Just be collaborative and folks here will embrace & help you make your own space to attack a problem that you know we'll all benefit from.

23.5 Questions?

- Shawn Garbett: shawn.garbett@vumc.org
- Günther Reznicek: g.reznicek@gmail.com
- Will Beasley: william-beasley@ou.edu